Reg. No. | | | | | | | | | | |

[13CE56]

FIFTH SEMESTER B.E EXAMINATION, DECEMBER - 2015

WATER SUPPLY ENGINEERING

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. Explain the importance and need for planned and protected water supply scheme **06**

   b. List the types of demand of water and explain it briefly. **08**

2. a. With the help of the following data estimate the population of a town for the year 2030 using incremental increase method **08**

Year	1950	1960	1970	1980	1990	2000	2010
Population	25000	27500	34100	41500	47050	54500	6100

   b. Define design period. Give the factors affecting design period of a water supply project. **06**

3. a. Explain the factors affecting the section and location of an intake structure. **06**

   b. With the help of neat diagram. Explain the working of centrifugal pump **08**

4. a. List the objectives of water quality management and explain the environmental significance of the following: **08**
   Nitrates and Fluorides

   b. Explain membrane filtration technique for determining the presence of bacteria in water. **06**

5. a. Define aeration. List the types of aerators and explain any one type with neat diagram **08**

   b. Explain the theory of sedimentation process in detail. **06**

6. a. Explain the working of rapid sand filters along with the method of cleaning with a neat sketch **07**

   b. Design the dimensions of a set of rapid gravity filters for treating water required for a population of 50,000 with a rate of supply of 180 liters per day per person, the filters are rated to work 5000 liters per hour per sq.m. Assume a peak factor of 1.8 by ignoring wash water requirements. **07**

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7. a. Explain the following briefly **08**

   i. Pre- chlorination

   ii. Post- chlorination

   iii. Break- point chlorination

   iv. Super –chlorination

   b. Briefly explain zeolite process for water softening. **06**

8. a. Give a comparison of continuous and intermittent system of water supply. **06**

   b. Explain Equivalent pipe method of water distribution system in detail. **08**